

Lecture 7: Impulse, Momentum and Collisions – Tutorial Sheet

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Use the Master Procedure: identify, sketch before and after, check that the system is isolated, predict, select the model, execute by components, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A head-on collision of a car and a truck

A 1000 kg car moving east at 15 m/s collides head on with a 3000 kg truck moving west at 5.0 m/s. They lock together. Find the velocity after the collision and the kinetic energy lost.

Solution. Take east as $+x$: $(1000)(15) + (3000)(-5.0) = (4000)v \Rightarrow v = 0$. Both stop. $K_{\text{before}} = \frac{1}{2}(1000)(15)^2 + \frac{1}{2}(3000)(5.0)^2 = 112.5 + 37.5 = 150 \text{ kJ}$; $K_{\text{after}} = 0$, so all 150 kJ is lost. *Check:* equal and opposite momenta (15 000 kg m/s each), so the total is zero.

A2. A two-dimensional completely inelastic collision

A 2.0 kg object moving east at 4.0 m/s collides with and sticks to a 3.0 kg object moving north at 2.0 m/s. Find the velocity after the collision.

Solution. x : $(2.0)(4.0) = (5.0)v_x \Rightarrow v_x = 1.6 \text{ m/s}$. y : $(3.0)(2.0) = (5.0)v_y \Rightarrow v_y = 1.2 \text{ m/s}$. Speed 2.0 m/s at $\tan^{-1}(1.2/1.6) = 37^\circ$ north of east. $K_{\text{before}} = 16 + 6 = 22 \text{ J}$, $K_{\text{after}} = \frac{1}{2}(5.0)(2.0)^2 = 10 \text{ J}$; inelastic.

B. Practice

B1. Impulse from a graph

The force on a 0.50 kg object initially at rest rises linearly from 0 to 100 N in 0.020 s, stays at 100 N for 0.030 s, then drops to zero. Find the impulse and the final speed.

B2. A bouncing ball

A 0.20 kg ball dropped from 2.0 m rebounds to 1.2 m. The contact with the floor lasts 10 ms. Find the impulse, the average force and the coefficient of restitution.

B3. Recoil

A 70 kg skater at rest on frictionless ice throws a 2.0 kg ball horizontally at 12 m/s. Find the skater's velocity.

B4. An elastic collision

A 2.0 kg block moving at 5.0 m/s collides elastically with a 3.0 kg block at rest. Find both velocities after the collision, and verify that kinetic energy is conserved.

B5. A glancing collision

A 1.0 kg ball moving east at 6.0 m/s hits a 2.0 kg ball at rest. The 1.0 kg ball leaves at 3.0 m/s, 60° north of east. Find the velocity of the 2.0 kg ball, and classify the collision.

B6. A ballistic pendulum

A 50 g bullet embeds in a 4.0 kg pendulum block, which rises 0.15 m. Find the bullet's speed.

B7. Challenge: the method is not given

A 0.15 kg ball moving at 10 m/s hits a 1.0 kg block at rest on a rough floor ($\mu_k = 0.30$), and rebounds at 4.0 m/s. How far does the block slide? What is the coefficient of restitution?

Answers

B1 $J = 4.0 \text{ N s}$; $v = 8.0 \text{ m/s}$

B2 $J = 2.2 \text{ N s}$ upward; $F_{\text{av}} = 2.2 \times 10^2 \text{ N}$; $e = \sqrt{1.2/2.0} = 0.77$

B3 -0.34 m/s (opposite to the ball)

B4 $v'_1 = -1.0 \text{ m/s}$, $v'_2 = +4.0 \text{ m/s}$; $K = 25 \text{ J}$ before and after

B5 2.6 m/s at 30° south of east; inelastic (K : 18 J $\rightarrow$ 11 J)

B6 $v = \frac{(M + m)\sqrt{2gh}}{m} = 1.4 \times 10^2 \text{ m/s}$

B7 block speed 2.1 m/s; slides 0.75 m; $e = 0.61$